

RAJ SHAH

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🔗 rajaxar.github.io/portfolio

Summary

Data scientist with 6+ years across causal inference, experimentation, and applied machine learning. MS in Data Science and Public Policy from Carnegie Mellon. Builds measurement frameworks end to end, from data pipelines and model development through deployment, and designs and analyzes experiments to quantify what actually changed. Applies quasi-experimental methods when randomizing is not feasible, and publishes data journalism explaining quantitative findings to general audiences.

Work Experience

BlackCloak AI

Jan 2026 - Present

Data Scientist / AI Engineer

Remote

- Designed location risk score for customer-facing product: five-level composite averaged from governance sources at country level plus event-driven regional adjustment that raises score.
- Built an adaptable insights platform so AI-generated summaries can be added across customer-facing services.
- Built internal tooling that automated the software development lifecycle, moving product pipelines from PRDs to shipped features in weeks rather than months.
- Developed plugins and skills for Codex, Cursor, and Claude Code that write specs, connect to Jira and Slack, and handle feature development and QA.

Aires

Jun 2025 - Jan 2026

Data Scientist

Pittsburgh, PA

- Developed Bayesian hierarchical models to estimate fixed and random effects driving customer survey outcomes.
- Modeled operational cash-flow time series with foundation forecasting models under limited historical data.
- Designed Python pipelines to ingest, clean, and analyze time-series data to identify causal structures.
- Designed analytical data models and reporting to support operational KPI dashboards and decision-making.

Allianz Life

May 2024 - Apr 2025

Data Science Intern

Minneapolis, MN

- Built an automated pipeline comparing 600+ permutations of models (BERT, LightGBM, XGBoost, scikit-learn) and feature-engineering methods, reaching 0.92 ROC-AUC while balancing accuracy, latency, and operational usability.
- Built a full-stack ML/NLP solution (tokenization, TF-IDF) for legal-document triage, translating classifier outputs into prioritization workflows that reduced review time by ~40% and saved ~\$20K annually.
- Prototyped a Cox proportional-hazards survival model to forecast policyholder lapses (exploratory analysis).
- Migrated dependencies for fraud-detection models worth \$4M+/quarter from legacy databases to the company standard using Python, SQL, and RStudio.

Ray Analytics (Startup)

Dec 2022 - Jul 2023

Founder / Chief Technology Officer

Washington, DC

- Trained, validated, and deployed XGBoost classifiers (94% ROC-AUC) on 12,000+ users from behavioral and engagement features to produce interpretable audience and influence scores.
- Founded and built a SaaS analytics platform for on-demand discourse analysis and policy-based influence identification, converting social/media engagement data into stakeholder-facing scores and reports.
- Used AWS API Gateway, Lambda, and React to build a backend and live dashboard serving 20+ customers.

Plainspoken Digital (Startup)

Mar 2022 - Jan 2023

Senior Software Engineer / Project Lead

Washington, DC

- Designed automated ETL workflows (PySpark) integrating cloud and on-prem data into one analytics layer.
- Led migration from legacy data systems to Palantir Foundry, improving operational reporting efficiency by 2x.
- Led Agile processes for a development team of 7 and managed intra-team project management.

Amazon Web Services (AWS)

Oct 2020 - Mar 2022

Software Development Engineer II (L5)

Crystal City, VA

- Performed big-data latency analysis with Pandas/Seaborn across a codebase receiving billions of daily requests.
- Led design and implementation of new features on the S3 Request Path.

- Implemented S3 Bucket Keys, reducing encryption-related cost overhead by up to 99% for enterprise customers.
- Onboarded and mentored several engineers across the S3 Object Processing org.

Data Science for Social Good

May 2020 - Aug 2020

Data Scientist

Jacksonville, FL

- Led a team of three to analyze data for 6,000+ adult learners served by literacy services in Florida.
- Conducted causal analysis to evaluate program efficacy and identify key drivers of learner attrition, reducing dropout rates by over 20%.

Booz Allen Hamilton

Jun 2019 - Aug 2019

Data Scientist

Washington, DC

- Programmed a Convolutional Neural Network with TensorFlow reaching 88% precision, using transfer learning.

Education

Carnegie Mellon University, Heinz College

Pittsburgh, PA

Master of Science, Data Science and Public Policy

May 2025

- **Coursework:** Applied Econometrics, Machine Learning, Decision Analytics, Optimization, Labor Economics, Introduction to Deep Learning, Sports Analytics, Geographic Information Systems, Telling Stories with Data

University of Florida, College of Engineering

Gainesville, FL

Bachelor of Science, Computer Science, Minor in Mathematics

May 2020

- **Coursework:** Database Systems, Introduction to Data Science, Numerical Analysis, Algorithmic Design, Linear Algebra

Personal Projects

Outcome Evaluation of Pittsburgh Re-Entry Services | *Causal inference*

Jan 2025 - Apr 2025

- Led causal inference on 2,800+ participants to evaluate impact of services on employment, earnings, and recidivism.
- Designed quasi-experiments and causal inference frameworks to measure program impact; built 15+ year panel datasets.
- Quantified 3x higher odds of employment for matched graduates.

Super Smash Bros. Melee Action-Value Analytics | *Production ML*

In Progress

- Built a win-probability value function predicting stock loss within an 8-second horizon; trained LightGBM on 10M+ frames at AUC 0.87, log-loss 0.27, Brier 0.082.
- Built a Bayesian hierarchical matchup model (PyMC) with crossed random effects over 60K+ games.
- Evaluated with GroupKFold by game to prevent frame leakage, plus calibration and probability-over-time plots.

Fund the Vote: Reducing Voter Wait Times | *Predict-then-optimize*

Aug 2024 - Dec 2024

- Predicted local voting turnout (Census data, GeoPandas, LassoCV, hierarchical regression) and optimized precinct resource allocation with a Gurobi model to reduce wait times for marginalized voters.

Data Journalism: Survivor and the Data of Diversity | *Visualization*

Jan 2026 - May 2026

- Built a published scrollytelling investigation into how Survivor casting and voting behavior shifted around diversity across all 49 seasons, with an interactive explorer for cast progression in any season.
- Quantified pre-merge targeting bias against players of color with confidence intervals, plus cast composition by game stage, age spread, and income diversity from BLS wage data.

Technical Skills

Causal Inference & Experimentation: Causal Inference (RCTs, Matching, Quasi-Experiments), Experiment Design, A/B Testing, Hypothesis Testing, Panel Data, Econometrics, Uncertainty Quantification, Forecasting, Survival Analysis

Statistical Modeling: Bayesian Hierarchical Modeling, Generalized Linear Models, Feature Engineering, Model Evaluation & Calibration, Optimization (Gurobi), Time Series

Machine Learning: Python (Pandas, scikit-learn, PyTorch, statsmodels, NumPy), R (Stan/brms, R Shiny), LightGBM, XGBoost, BERT, NLP, Model Deployment

Data Engineering & Programming: SQL, ETL/ELT, Spark/PySpark, Production Data Pipelines, Palantir Foundry, AWS (Lambda, S3, Bedrock), GCP (Cloud Run, Pub/Sub), Git, CI/CD

Metrics, Visualization & Communication: Metric & KPI Design, Dashboarding & Self-Serve Tooling, Tableau, Power BI, Streamlit, Published Data Journalism, Stakeholder Reporting, Technical Mentoring